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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	TMP
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 - Quality Education
Project / Solution Name	Noka
Name (Leader)	Rohan
Email & Phone (Leader)	dr.rohansaravanan@gmail.com 0175522805
University Affiliation(s)	APU
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is changing the tasks within occupations, leaving students, graduates and working professionals with a practical question: "What should I learn and build now to remain relevant?" The World Economic Forum's Future of Jobs Report 2025 estimates that 39% of workers' existing skills will change or become outdated between 2025 and 2030, while 59 out of every 100 workers will require training. These are global employer expectations, rather than measurements of individual career outcomes.

For someone planning a career transition, the information needed to act is fragmented across job advertisements, occupational frameworks, AI announcements and learning platforms. Users must manually identify recurring requirements, assess their own gaps and decide which projects would demonstrate relevant capabilities. Generic recommendations often overlook their existing experience, chosen destination and available learning time.

This creates a gap between awareness of workforce change and an achievable response. People with limited access to career guidance may struggle to prioritise learning, while employers face applicants whose training does not translate clearly into workplace tasks.

Noka addresses this decision-making gap, initially focusing on university students, early-career professionals and workers exploring transitions. It connects career evidence to personalised action while helping users understand how their skills and potential work relate to sustainable development.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Noka is an AI-powered career and reskilling workspace that connects a person's current capabilities to a future destination, real job requirements and a practical learning project.

Users begin by submitting experience or reviewing their skills. Noka structures this information and compares confirmed skills with occupational requirements. Its interactive Career Universe makes transferable capabilities and gaps visible in a fun way, helping users choose and save a target role.

For that destination, Noka researches public job descriptions through Tavily or accepts a manually pasted description. Generative AI extracts requirements into a structured format for user review, then proposes a project and milestones informed by the user's starting skills and weekly learning commitment. This turns "learn AI" into a more concrete objective, such as building and evaluating a cloud-based AI application.

Intelligence Radar presents source-linked developments and career signals. Relevant evidence can inform proposed pathway updates, which users may accept while retaining existing progress. The World Dial explores alternative future scenarios, while My Impact explains supported connections between roles, skills and UN Sustainable Development Goals.

The architecture combines GenAI APIs, structured data validation, relational database design and Replit deployment. AI performs extraction and interpretation; application logic calculates skill coverage, manages ownership and preserves progress.

The prototype uses React, FastAPI, PostgreSQL, OpenAI and Tavily, with Replit deployment. Reviewed evidence and synthetic demonstration scenarios are explicitly labelled, allowing judges to explore the full experience without confusing illustrative assumptions with observed labour-market findings.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	**Technology and purpose**
User interface	React, TypeScript and Vite deliver the career workspace; React Flow visualises connections among skills, roles and opportunities.
Application API	Python and FastAPI coordinate profile analysis, career matching, research and pathway updates.
Generative AI	OpenAI APIs extract structured skills and job requirements, explain evidence and propose personalised projects.
Public research	Tavily discovers public job descriptions and relevant developments. Private resumes are excluded from search requests.
Data and validation	PostgreSQL, SQLAlchemy, Alembic and Pydantic manage persistent records, database migrations and validated data contracts.

Evidence sources	O*NET, public labour-market publications and official UN SDG information provide occupational and sustainability context.
Hosting	**Replit** is deployment platform, serving the compiled frontend and FastAPI backend through one application service.
Quality assurance	pytest, Vitest and React Testing Library test API behaviour, user interactions and progress preservation.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathize: We recognized a shared challenge among students and working adults: uncertainty about how AI will reshape careers and which skills they should develop to remain competitive.

Define: We identified the need for an accessible, personalized platform that connects emerging career opportunities, individual skill gaps, and actionable reskilling pathways.

Ideate: We conceptualized Noka as an AI-powered career companion integrating workforce intelligence, future scenario exploration, career matching, and personalized learning roadmaps.

Prototype: We developed and deployed a functional Noka application on Replit, featuring interactive career exploration, skill profiling, AI-assisted insights, and reskilling pathways.

Test: In Phase 2, we plan to pilot Noka with 2 students and 1 early-career professional, evaluating usability, task completion, recommendation relevance, and roadmap practicality to guide improvements.

Deployment Strategy: With Noka already live on Replit, Phase 2 will focus on production readiness through secure user authentication, persistent individual accounts, protected profile data, and reliable returning-user experiences. We will additionally strengthen database persistence, automated testing,

API security, background research reliability, and deployment monitoring to support real-world adoption.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4—increase relevant technical and vocational skills among youth and adults for employment, decent jobs and entrepreneurship. Noka supports this through requirement-based projects and reskilling pathways.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published context: WEF estimates that 59% of workers will require training by 2030, illustrating the scale of the reskilling challenge. WEF Skills Outlook Prototype evaluation baseline: Use 40 minutes per manually prepared career plan as a provisional planning assumption, not a measured finding.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction in median time to produce a qualifying career plan

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	25% reduction in career-planning task completion time compared to manual research, measured through Phase 2 usability testing with students and working professionals.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Model & Version	Source / Provider	Usage	License / Terms
GPT-4.1-mini	OpenAI	Skill extraction from resumes or manually entered text	OpenAI API terms
gpt-6-sol	OpenAI	Pre-trained; Research analysis, pathway/project proposals and evidence explanations. It also used by Guide agent to select tools and retrieve application evidence	OpenAI API terms

Persistence baseline	Noka implementation	Self-trained/evaluated from historical series as a forecasting baseline **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	Noka implementation; no separate model licence
Ridge Regression, alpha=10	scikit-learn	Self-trained by sector and forecast horizon using historical US sector data **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	scikit-learn BSD-3-Clause
XGBoost, 80 trees, depth 3	XGBoost	Self-trained forecasting candidate using historical US sector data **Note:** The forecasting module was initially experimented with but later removed, as its results did not provide sufficient value or meaningful improvement to the overall system's performance.	XGBoost Apache-2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Rohan

[Signature]

[Date] 24/10/2026